2026.07.11--所有实验结果指令汇总

数据集：arxiv

```
for dim in 32 64 128 256 384 512 768; do echo "=== GCN arxiv dim=$dim ===" && python train_unified_methods.py --method gcn --dataset arxiv --root ./data --mode full --gpu-id 0 --infer-device auto --hidden-dim ${dim} --num-layers 2 --dropout 0.5 --supervised-epochs 500 --supervised-lr 0.01 --supervised-weight-decay 0.0005 --supervised-early-stop --output-dir ./results && echo "=== GCN arxiv dim=$dim done ==="; done

for dim in 32 64 128 256 384 512 768; do python train_unified_methods.py --method dgi --dataset arxiv --mode full --gpu-id 0 --hidden-dim ${dim} --num-layers 2 --dropout 0.0 --pretrain-epochs 300 --pretrain-lr 0.001 --pretrain-weight-decay 0.0 --cls-epochs 200 --cls-lr 0.01 --cls-weight-decay 0.0 --output-dir dgi_arxiv_dims; done

python train_unified_methods.py --method dgi_mrl --dataset arxiv --mode full --gpu-id auto --hidden-dim 768 --num-layers 2 --dropout 0.0 --pretrain-epochs 300 --pretrain-lr 0.001 --pretrain-weight-decay 0.0 --mrl-dims 32,64,128,256,384,512,768 --output-dir dgi_mrl_arxiv_h768

 for dim in 32 64 128 256 384 512 768; do python train_unified_methods.py --method ccassg --dataset arxiv --mode full --gpu-id 0 --hidden-dim ${dim} --num-layers 2 --dropout 0.0 --pretrain-epochs 100 --pretrain-lr 0.001 --pretrain-weight-decay 0.0 --lam 0.001 --p-feat-mask-1 0.0 --p-edge-drop-1 0.5 --p-feat-mask-2 0.0 --p-edge-drop-2 0.5  --output-dir ccassg_arxiv_dims; done

python train_unified_methods.py --method ccassg_mrl --dataset arxiv --mode full --gpu-id 1 --hidden-dim 768 --num-layers 2 --dropout 0.0 --pretrain-epochs 100 --pretrain-lr 0.001 --pretrain-weight-decay 0.0 --lam 0.001 --p-feat-mask-1 0.0 --p-edge-drop-1 0.5 --p-feat-mask-2 0.0 --p-edge-drop-2 0.5 --mrl-dims 32,64,128,256,384,512,768 --output-dir ccassg_mrl_arxiv_h768

for dim in 32 64 128 256 384 512 768;do python train_unified_methods.py --method grace --dataset arxiv  --mode neighbor --gpu-id auto --hidden-dim ${dim} --num-layers 3 --dropout 0.0 --proj-dim ${dim} --tau 0.1 --pretrain-epochs 200 --pretrain-lr 0.001 --pretrain-weight-decay 0.00001 --batch-size 16384 --num-neighbors 10 10 10 --eval-batch-size 256 --eval-num-neighbors -1 -1 -1 --p-feat-mask-1 0.0 --p-edge-drop-1 0.4 --p-feat-mask-2 0.0 --p-edge-drop-2 0.4 --output-dir grace_arxiv_dims;done

python train_unified_methods.py --method grace_mrl --dataset arxiv  --mode neighbor --gpu-id auto --hidden-dim 768 --proj-dim 768 --num-layers 3 --dropout 0.0 --tau 0.1 --pretrain-epochs 200 --pretrain-lr 0.001 --
```

```
pretrain-weight-decay 0.00001 --batch-size 16384 --num-neighbors 10 10 10
--eval-batch-size 256 --eval-num-neighbors -1 -1 -1 --p-feat-mask-1 0.0 --
p-edge-drop-1 0.4 --p-feat-mask-2 0.0 --p-edge-drop-2 0.4 --mrl-dims
32,64,128,256,384,512,768 --output-dir grace_mrl_arxiv_h768

python train_unified_methods.py --method grace_ml --dataset arxiv --mode
neighbor --gpu-id 0 --infer-device 0 --hidden-dim 768 --num-layers 3 --
dropout 0.0 --proj-dim 768 --tau 0.1 --grace-only-epochs 200 --grace-ml-
epochs 200 --pretrain-lr 0.001 --grace-ml-lr 0.0001 --pretrain-weight-
decay 0.00001 --batch-size 4096 --num-neighbors 10 10 10 --eval-batch-size
256 --eval-num-neighbors -1 -1 -1 --p-feat-mask-1 0.0 --p-edge-drop-1 0.4
--p-feat-mask-2 0.0 --p-edge-drop-2 0.4 --mrl-dims
32,64,128,256,384,512,768 --mrl-weight 1.0 --ml-weight 0.1 --ml-module ml2
--output-dir grace_ml_arxiv_v4_ml2

python train_unified_methods.py --method csne --dataset arxiv --root
./data --mode neighbor --gpu-id 1 --infer-device 1 --hidden-dim 768 --num-
layers 3 --dropout 0.0 --proj-dim 768 --tau 0.1 --grace-only-epochs 200 --
grace-ml-epochs 200 --pretrain-lr 0.001 --grace-ml-lr 0.00001 --pretrain-
weight-decay 0.00001 --batch-size 4096 --num-neighbors 10 10 10  --eval-
batch-size 256 --eval-num-neighbors -1 -1  -1 --p-feat-mask-1 0.0 --p-
edge-drop-1 0.4 --p-feat-mask-2 0.0 --p-edge-drop-2 0.4 --mrl-dims
32,64,128,256,384,512,768 --mrl-weight 1.0 --ml-weight 1.0 --ml-module ml2
--hpem-beta-init 0.1 --hpem-tau-0 0.1 --output-dir csne_arxiv_v1
```

数据集：products

```
for dim in  512 768; do echo "=== GCN dim=$dim ===" && python
train_unified_methods.py --method gcn --dataset products --mode neighbor -
-gpu-id auto --infer-device auto --hidden-dim ${dim} --num-layers 2 --
dropout 0.5 --supervised-epochs 100 --supervised-lr 0.01 --supervised-
weight-decay 0.0005 --supervised-early-stop --batch-size 1024 --num-
neighbors 10 10 --eval-batch-size 1024 --eval-num-neighbors 10 10 --
output-dir ./results && echo "=== GCN dim=$dim done ==="; done

for dim in 32 64 128 256 384 512 768; do echo "=== DGI dim=$dim ===" &&
python train_unified_methods.py --method dgi --dataset products --root
./data --mode neighbor --gpu-id 1 --infer-device auto --hidden-dim $dim --
num-layers 2 --dropout 0.0 --pretrain-epochs 10 --pretrain-lr 0.001 --
pretrain-weight-decay 0.0 --batch-size 4096 --num-neighbors 10 10 --eval-
batch-size 4096 --eval-num-neighbors 10 10 --cls-epochs 200 --cls-lr 0.01
--cls-batch-size 8192 --cls-early-stop --output-dir ./results && echo "===
DGI dim=$dim done ==="; done

echo "=== DGI+MRL hidden_dim=768 ===" && python train_unified_methods.py -
-method dgi_mrl --dataset products --mode neighbor --gpu-id 1 --infer-
device auto --hidden-dim 768 --num-layers 2 --dropout 0.0 --pretrain-
epochs 10 --pretrain-lr 0.001 --pretrain-weight-decay 0.0 --batch-size
```

```
4096 --num-neighbors 10 10 --eval-batch-size 4096 --eval-num-neighbors 10
10 --mrl-dims 32,64,128,256,384,512,768 --cls-epochs 200 --cls-lr 0.01 --
cls-batch-size 8192 --cls-early-stop --output-dir ./results && echo "===
DGI+MRL hidden_dim=768 done ==="

for dim in 32 64 128 256 384 512 768; do echo "=== CCA-SSG dim=$dim ==="
&& python train_unified_methods.py --method ccassg --dataset products --
mode neighbor --gpu-id 1 --infer-device auto --hidden-dim "${dim}" --num-
layers 2 --dropout 0.0 --pretrain-epochs 10 --pretrain-lr 0.001 --
pretrain-weight-decay 0.0 --batch-size 4096 --num-neighbors 10 10 --eval-
batch-size 4096 --eval-num-neighbors 10 10 --lam 0.001 --p-feat-mask-1 0.0
--p-edge-drop-1 0.5 --p-feat-mask-2 0.0 --p-edge-drop-2 0.5 --cls-epochs
200 --cls-lr 0.01 --cls-batch-size 8192 --cls-early-stop --output-dir
./results && echo "=== CCA-SSG dim=$dim done ==="; done

echo "=== CCA-SSG+MRL hidden_dim=768 ===" && python
train_unified_methods.py --method ccassg_mrl --dataset products --root
./data --mode neighbor --gpu-id 0 --infer-device auto --hidden-dim 768 --
num-layers 2 --dropout 0.0 --pretrain-epochs 10 --pretrain-lr 0.001 --
pretrain-weight-decay 0.0 --batch-size 4096 --num-neighbors 10 10 --eval-
batch-size 4096 --eval-num-neighbors 10 10 --lam 0.001 --p-feat-mask-1 0.0
--p-edge-drop-1 0.5 --p-feat-mask-2 0.0 --p-edge-drop-2 0.5 --mrl-dims
32,64,128,256,384,512,768 --cls-epochs 200 --cls-lr 0.01 --cls-batch-size
8192 --cls-early-stop --output-dir ./results && echo "=== CCA-SSG+MRL
hidden_dim=768 done ==="


for dim in 32 64 128 256 384 512 768; do echo "=== GRACE dim=$dim ===" &&
python train_unified_methods.py --method grace --dataset products --mode
neighbor --gpu-id 0 --infer-device 1 --hidden-dim ${dim} --proj-dim ${dim}
--num-layers 2 --dropout 0.0 --tau 0.1 --pretrain-epochs 20 --pretrain-lr
0.001 --pretrain-weight-decay 0.00001 --batch-size 1024 --num-neighbors 10
10 --eval-batch-size 4096 --eval-num-neighbors 10 10 --p-feat-mask-1 0.0 -
-p-edge-drop-1 0.5 --p-feat-mask-2 0.0 --p-edge-drop-2 0.5 --cls-epochs
200 --cls-lr 0.01 --cls-batch-size 8192 --cls-early-stop --output-dir
./results && echo "=== GRACE dim=$dim done ==="; done

export PYTORCH_CUDA_ALLOC_CONF=expandable_segments:True && echo "===
GRACE+MRL products hidden_dim=768 ===" && python train_unified_methods.py
--method grace_mrl --dataset products --mode neighbor --gpu-id 1 --infer-
device auto --hidden-dim 768 --proj-dim 768 --num-layers 2 --dropout 0.0 -
-tau 0.1 --pretrain-epochs 40 --pretrain-lr 0.001 --pretrain-weight-decay
0.00001 --batch-size 1024 --num-neighbors 10 10 --eval-batch-size 4096 --
eval-num-neighbors 10 10 --p-feat-mask-1 0.0 --p-edge-drop-1 0.5 --p-feat-
mask-2 0.0 --p-edge-drop-2 0.5 --mrl-dims 32,64,128,256,384,512,768 --cls-
epochs 200 --cls-lr 0.01 --cls-batch-size 8192 --cls-early-stop --output-
dir ./results && echo "=== GRACE+MRL products hidden_dim=768 done ==="


python train_unified_methods.py --method grace_ml --dataset products --
```

```
root ./data --mode neighbor --gpu-id 0 --infer-device 0 --hidden-dim 768 --num-layers 2 --dropout 0.0 --proj-dim 768 --tau 0.1 --grace-only-epochs 10 --grace-ml-epochs 30 --pretrain-lr 0.001 --grace-ml-lr 0.00001 --pretrain-weight-decay 0.00001 --batch-size 1024 --num-neighbors 10 10 --eval-batch-size 4096 --eval-num-neighbors 10 10 --p-feat-mask-1 0.0 --p-edge-drop-1 0.5 --p-feat-mask-2 0.0 --p-edge-drop-2 0.5 --mrl-dims 32,64,128,256,384,512,768 --mrl-weight 1.0 --ml-weight 0.1 --ml-module ml2 --output-dir grace_ml_products_v4_ml2

python train_unified_methods.py --method csne --dataset products --root ./data --mode neighbor --gpu-id 1 --infer-device 1 --hidden-dim 768 --num-layers 2 --dropout 0.0 --proj-dim 768 --tau 0.1 --grace-only-epochs 10 --grace-ml-epochs 30--pretrain-lr 0.001 --grace-ml-lr 0.00001 --pretrain-weight-decay 0.00001 --batch-size 1024 --num-neighbors 10 10 --eval-batch-size 4096 --eval-num-neighbors 10 10 --p-feat-mask-1 0.0 --p-edge-drop-1 0.5 --p-feat-mask-2 0.0 --p-edge-drop-2 0.5 --mrl-dims 32,64,128,256,384,512,768 --mrl-weight 1.0 --ml-weight 0.1 --ml-module ml2 --hpem-beta-init 0.1 --hpem-tau-0 0.1 --output-dir csne_products_v1
```

数据集：MAG

```
for dim in 32 64 128 256 384 512 768; do echo "=== GCN MAG dim=$dim ===" && python train_unified_methods.py --method gcn --dataset mag --root ./data --mode neighbor --gpu-id 0 --infer-device auto --num-layers 2 --hidden-dim ${dim} --dropout 0.5 --supervised-epochs 100 --supervised-lr 0.01 --supervised-weight-decay 0.0005 --supervised-early-stop --batch-size 1024 --num-neighbors 10 10 --eval-batch-size 4096 --eval-num-neighbors 10 10 --output-dir ./results && echo "=== GCN MAG dim=$dim done ==="; done

for dim in 256 384; do echo "=== GCN MAG dim=$dim ===" && python train_unified_methods.py --method gcn --dataset mag --root ./data --mode neighbor --gpu-id 0 --infer-device auto --num-layers 2 --hidden-dim ${dim} --dropout 0.5 --supervised-epochs 100 --supervised-lr 0.01 --supervised-weight-decay 0.0005 --supervised-early-stop --batch-size 1024 --num-neighbors 10 10 --eval-batch-size 4096 --eval-num-neighbors 10 10 --output-dir ./results && echo "=== GCN MAG dim=$dim done ==="; done

  export PYTORCH_CUDA_ALLOC_CONF=expandable_segments:True && for dim in 32 64 128 256 384 512 768; do echo "=== DGI MAG dim=$dim ===" && python train_unified_methods.py --method dgi --dataset mag --root ./data --mode neighbor --gpu-id 1 --infer-device 1 --num-layers 2 --hidden-dim ${dim} --dropout 0.0 --pretrain-epochs 50 --pretrain-lr 0.001 --pretrain-weight-decay 0.0 --batch-size 1024 --num-neighbors 10 10 --eval-batch-size 4096 --eval-num-neighbors 10 10 --cls-epochs 200 --cls-lr 0.01 --cls-batch-size 8192 --cls-early-stop --output-dir ./results && echo "=== DGI MAG dim=$dim done ==="; done
```

```
  export PYTORCH_CUDA_ALLOC_CONF=expandable_segments:True && echo "===
DGI+MRL MAG hidden_dim=768 ===" && python train_unified_methods.py --
method dgi_mrl --dataset mag --root ./data --mode neighbor --gpu-id 0 --
infer-device auto --num-layers 2 --hidden-dim 768 --dropout 0.0 --mrl-dims
32,64,128,256,384,512,768 --pretrain-epochs 50 --pretrain-lr 0.001 --
pretrain-weight-decay 0.0 --batch-size 1024 --num-neighbors 10 10 --eval-
batch-size 4096 --eval-num-neighbors 10 10 --cls-epochs 200 --cls-lr 0.01
--cls-batch-size 8192 --cls-early-stop --output-dir ./results && echo "===
DGI+MRL MAG hidden_dim=768 done ==="

export PYTORCH_CUDA_ALLOC_CONF=expandable_segments:True && for dim in 32
64 128 256 384 512 768; do echo "=== CCA-SSG MAG hidden_dim=${dim} ===" &&
python train_unified_methods.py --method ccassg --dataset mag --mode
neighbor --gpu-id 0 --infer-device 0 --num-layers 2 --hidden-dim ${dim} --
dropout 0.0 --lam 1e-3 --p-feat-mask-1 0.0 --p-edge-drop-1 0.5 --p-feat-
mask-2 0.0 --p-edge-drop-2 0.5 --pretrain-epochs 50 --pretrain-lr 0.001 --
pretrain-weight-decay 0.0 --batch-size 4096 --num-neighbors 10 10 --eval-
batch-size 4096 --eval-num-neighbors 10 10 --cls-epochs 200 --cls-lr 0.01
--cls-weight-decay 0.0 --cls-batch-size 8192 --cls-early-stop --output-dir
./results && echo "=== CCA-SSG MAG hidden_dim=${dim} done ==="; done


export PYTORCH_CUDA_ALLOC_CONF=expandable_segments:True && echo "=== CCA-
SSG+MRL MAG hidden_dim=768 ===" && python train_unified_methods.py --
method ccassg_mrl --dataset mag  --mode neighbor --gpu-id 0 --infer-device
0 --num-layers 2 --hidden-dim 768 --dropout 0.0 --lam 1e-3 --p-feat-mask-1
0.0 --p-edge-drop-1 0.5 --p-feat-mask-2 0.0 --p-edge-drop-2 0.5 --mrl-dims
"32,64,128,256,384,512,768" --mrl-weight 1.0 --pretrain-epochs 50 --
pretrain-lr 0.001 --pretrain-weight-decay 0.0 --batch-size 4096 --num-
neighbors 10 10 --eval-batch-size 4096 --eval-num-neighbors 10 10 --cls-
epochs 200 --cls-lr 0.01 --cls-weight-decay 0.0 --cls-batch-size 8192 --
cls-early-stop --output-dir ./results && echo "=== CCA-SSG+MRL MAG
hidden_dim=768 done ==="

  export PYTORCH_CUDA_ALLOC_CONF=expandable_segments:True && for dim in
32 64 128 256 384 512 768; do echo "=== GRACE MAG hidden_dim=${dim} ==="
&& python train_unified_methods.py --method grace --dataset mag --root
./data --mode neighbor --gpu-id 1 --infer-device auto --num-layers 2 --
hidden-dim ${dim} --proj-dim ${dim} --dropout 0.0 --tau 0.1 --p-feat-mask-
1 0.0 --p-edge-drop-1 0.6 --p-feat-mask-2 0.0 --p-edge-drop-2 0.6 --
pretrain-epochs 50 --pretrain-lr 1e-3 --pretrain-weight-decay 1e-5 --
batch-size 4096 --num-neighbors 10 10 --eval-batch-size 256 --eval-num-
neighbors 10 10 --cls-epochs 200 --cls-lr 0.01 --cls-weight-decay 0.0 --
cls-batch-size 8192 --cls-early-stop --output-dir ./results && echo "===
GRACE MAG hidden_dim=${dim} done ==="; done


export PYTORCH_CUDA_ALLOC_CONF=expandable_segments:True && echo "===
GRACE+MRL MAG hidden_dim=768 ===" && python train_unified_methods.py --
method grace_mrl --dataset mag --root ./data --mode neighbor --gpu-id 1 --
```

```
infer-device 1 --num-layers 2 --hidden-dim 768 --proj-dim 768 --dropout
0.0 --tau 0.1 --p-feat-mask-1 0.0 --p-edge-drop-1 0.6 --p-feat-mask-2 0.0
--p-edge-drop-2 0.6 --mrl-dims "32,64,128,256,384,512,768" --mrl-weight
1.0 --pretrain-epochs 50 --pretrain-lr 1e-3 --pretrain-weight-decay 1e-5 -
-batch-size 4096 --num-neighbors 10 10 --eval-batch-size 256 --eval-num-
neighbors 10 10 --cls-epochs 200 --cls-lr 0.01 --cls-weight-decay 0.0 --
cls-batch-size 8192 --cls-early-stop --output-dir ./results && echo "===
GRACE+MRL MAG hidden_dim=768 done ==="

python train_unified_methods.py --method grace_ml --dataset mag --root
./data --mode neighbor --gpu-id 1 --infer-device 1 --hidden-dim 768 --num-
layers 2 --dropout 0.0 --proj-dim 768 --tau 0.1 --grace-only-epochs 50 --
grace-ml-epochs 50 --pretrain-lr 0.001 --grace-ml-lr 0.00001 --pretrain-
weight-decay 0.00001 --batch-size 4096 --num-neighbors 10 10 --eval-batch-
size 256 --eval-num-neighbors 10 10 --p-feat-mask-1 0.0 --p-edge-drop-1
0.6 --p-feat-mask-2 0.0 --p-edge-drop-2 0.6 --mrl-dims
32,64,128,256,384,512,768 --mrl-weight 1.0 --ml-weight 0.1 --ml-module ml2
--output-dir grace_ml_mag_v4_ml2

python train_unified_methods.py --method csne --dataset mag --root ./data
--mode neighbor --gpu-id 1 --infer-device 1 --hidden-dim 768 --num-layers
2 --dropout 0.0 --proj-dim 768 --tau 0.1 --grace-only-epochs 50 --grace-
ml-epochs 50 --pretrain-lr 0.001 --grace-ml-lr 0.00001 --pretrain-weight-
decay 0.00001 --batch-size 4096 --num-neighbors 10 10 --eval-batch-size
256 --eval-num-neighbors 10 10 --p-feat-mask-1 0.0 --p-edge-drop-1 0.6 --
p-feat-mask-2 0.0 --p-edge-drop-2 0.6 --mrl-dims 32,64,128,256,384,512,768
--mrl-weight 1.0 --ml-weight 1.0 --ml-module ml2 --hpem-beta-init 0.1 --
hpem-tau-0 0.1 --output-dir csne_mag_v1
```